Avni Bhardwaj

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EDUCATION

Narsee Monjee Institute of Management Studies (NMIMS),
B.TECH. in Computer Engineering, GPA: 3.37 / 4.0 *Aug 2022 - Aug 2026*

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL (MySQL, PostgreSQL), HTML/CSS

Mobile and Web Development: React.js, Android Native (Java, Android Studio), Firebase, Node.js, REST APIs, FastAPI, Component-Based Architecture, Microservices, JSON, OOP

Systems & Cloud: Linux/Unix, AWS (S3, EC2, ECS, Aurora), Google Cloud, Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Git,

Data Engineering: ETL Pipeline Design, Data Modeling, Data Warehousing, Kafka/Redpanda, Apache Flink, Redis, MinIO, Grafana, MySQL, Qdrant (Vector DB)

Productivity & Analytics Tools: Microsoft Excel, Word, PowerPoint, Tableau

AI / Machine Learning: Generative AI, LLMs, Agentic AI, RAG, NLP, LangChain, LlamaIndex, Prompt Engineering, PyTorch, Scikit-Learn, NumPy, Pandas

WORK EXPERIENCES

Flight Physics Capabilities Intern — Airbus *Bangalore, India*
Python, FastAPI, LlamaIndex, Qdrant, vLLM, Docker, Jenkins, NetworkX, PySide6, Linux *Jan 2026 – Jul 2026*

- Architected FP-BOT, a Dockerized RAG pipeline deployed on Linux via Jenkins CI/CD, using LlamaIndex, vLLM, and Qdrant to index documentation from 30+ tools for natural-language querying via FastAPI.
- Architected AirSimuPy's software architecture from scratch as a proof of concept for the Flight Physics team, developing a Python/PySide6 block-diagram simulation engine using NetworkX topological sorting and a NumPy-backed execution kernel, reducing team reliance on MATLAB Simulink.
- Validated a Python load-sorting engine against a legacy Fortran solver on Airbus's Linux compute cluster.Created test scenarios to identify race conditions, data-shape errors, and multiprocessing serialization issues. The results showed numerical parity with the Fortran benchmark across millions of data points.
- Built a multi-threaded Tkinter test-automation tool with SSH batch dispatch and Regex-based path rewriting, cutting a 10-file validation test cycle from ~20 minutes to under 60 seconds (95%+ reduction).

Software Developer Intern — Hiteshi Infotech *Indore, India*
React.js, JavaScript, HTML/CSS, Node.js *Jun 2024 – Jul 2024*

- Contributed to the company's production website using React.js, building and integrating reusable UI components with HTML/CSS and JavaScript.
- Learned and applied component-based frontend architecture and state management patterns, and worked with Node.js for backend-connected features.

PROJECTS

Personalized AI Financial Advisor | *Python, Unsloth, LangChain, Ollama, PyTorch* Apr 2025

- Built a data pipeline to clean, transform, and load 70k+ financial records (Finance-Alpaca, FiQA) for fine-tuning Llama-3 (8B) via PyTorch-based Unsloth.
- Architected an Agentic RAG system (LangChain ReAct) ingesting real-time stock APIs (yfinance) and Kaggle CSV data into a Streamlit UI for automated financial recommendations.

MapMyNotes — AI-Powered NLP Study Companion | *Python, Streamlit, Google Gemini API, D3.js* Nov 2025

- Built an NLP pipeline that converts uploaded notes, PDFs, and PPTX files into structured, interactive D3.js mind maps, using Gemini to extract hierarchical topics/subtopics at unlimited depth.
- Added PDF and PowerPoint parsing using PyMuPDF and python-pptx and built a keyword-extraction pipeline to generate the mind-map structure. Added AI-generated explanations on hover, with both simple and technical descriptions for each node.
- Developed auto-generated flashcards and quizzes from summarized content in a Streamlit UI, with on-demand regeneration and PNG export of the generated mind map.

Driving License Application Portal (Full-Stack) | *React.js, FastAPI, PostgreSQL, Razorpay API, Tailwind CSS* Mar 2026

- Built a full-stack citizen service portal for age-gated license applications, with role-based access control (RBAC) separating the applicant flow from the admin verification dashboard.
- Developed a persistent multi-step application wizard using React state and localStorage, letting users resume partially completed forms across sessions.
- Integrated Razorpay for application fees, with FastAPI webhook endpoints that verify payment events and update PostgreSQL application records automatically.
- Designed a normalized PostgreSQL schema linking users, payment states, and document storage URLs, using transactions to keep CRUD operations consistent.

IoT Smart Home Automation | *ESP32, C++, DHT11, MQ-2, PIR, LDR, LCD, Blynk* Nov 2025

- Developed an ESP32-based smart home automation system for IOT project which integrates environmental, gas, light, and motion sensors for real-time monitoring and automated control.
- Used sensor readings to control appliances through relay switching, with the LCD displaying system status and buzzer/LED alerts for events.

Confidential Deep Learning Research (Under Review, Springer Journal) | *PyTorch, DenseNet201, SE-ResNet50* 2025

- Architected an ensemble deep learning framework (4-author team) integrating **4 advanced models** for automated bone-age assessment from pediatric hand radiographs. to optimize image classification and diagnostic accuracy.
- Implemented image preprocessing, augmentation, model training, hyperparameter tuning, and ensemble evaluation using the RSNA Pediatric Bone Age dataset.

Hybrid Face-Iris Biometric Authentication Research (Under Review for Publication at IEEE) | *PyTorch, Cross-Attention Transformers, ResNet, VGG16*

- Co-authored a research paper (4-author team) proposing and evaluating three novel cross-attention deep learning architectures for multimodal face-iris biometric verification, **achieving up to 99.83% accuracy in evaluation.**
- Contributed to model design, experimentation, and benchmarking across public biometric datasets; full architectural and methodological details withheld pending publication.

CERTIFICATIONS & ACHIEVEMENTS

- Fast Prototyping of GenAI Apps with Streamlit — DeepLearning.AI
- Generative AI with Large Language Models: Transformers in Practice — DeepLearning.AI
- AWS Academy Graduate — Cloud Architecting, AWS Academy
- Google Cloud Fundamentals — Google Cloud Study Jams
- Amazon ML Summer School 2024 — Selected Participant
- NVIDIA Certified Associate — Generative AI LLMs (NCA-GENL) Cert Prep — LinkedIn Learning
- 2× Winner — Hitnext Challenge, GDSC NMIMS Indore
- Finalist — IIM Indore Chaitanya Leadership Event, 2024
- Finalist — Kavach Hackathon, Government of India Cybersecurity Hackathon, 2023

LEADERSHIP

Lead — Google Developer Student Clubs (GDSC), NMIMS Indore *2023 – 2024*

Led the campus GDSC chapter and a community of **300+ students**, organizing **10+ technical workshops** on emerging technologies, including AI, cloud computing, and developer technologies.

Led two Google Cloud certification initiatives that helped 200+ students earn Google Cloud and GenAI credentials through free certification programs.

Achieved Tier 1 recognition among GDSC chapters for chapter performance and student engagement, earning **recognition from Google** for successful execution of technical initiatives and community programs.

Organizer — Witty Hacks Hackathon *(Central India's Largest Hackathon)* *2024*

Co-organized as a main Head for Witty Hacks, coordinating logistics, operations, sponsor and finance relations for 3000+ participants across 100 teams hosted at NMIMS indore for a 3 day event.

Creative Member — Turing Club, NMIMS Indore *2022 – 2023*

Designed UI/UX mockups and promotional posters for club events and campaigns as part of the internal creative/design team.